

Lecture 4

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Jacaranda Plant (Training)

- Determine the Views
- Position the Views
- Style the Views

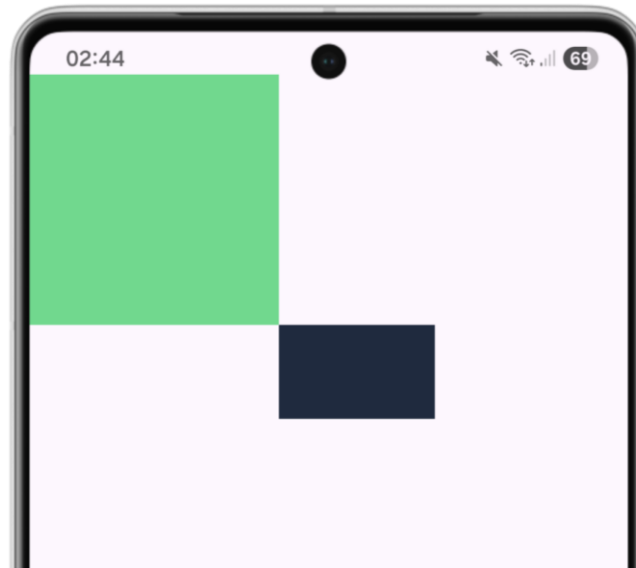


Break

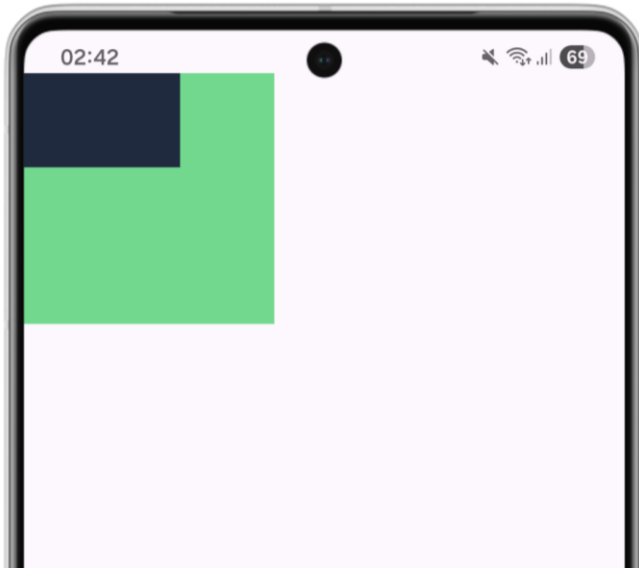


Training

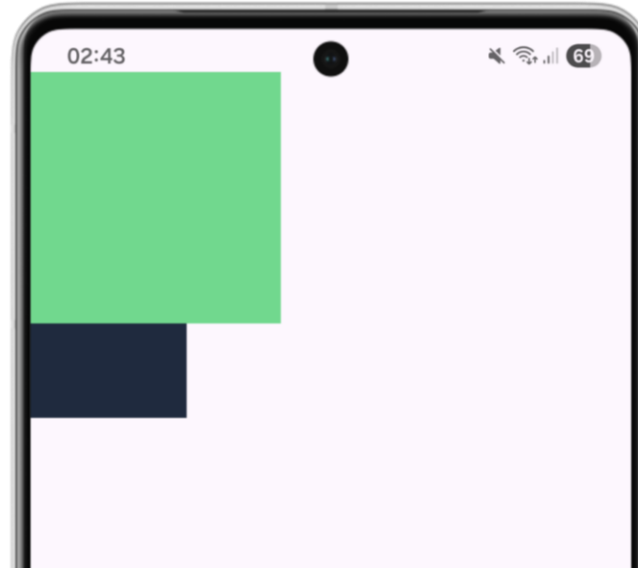
How can you build the following UI using a LinearLayout?



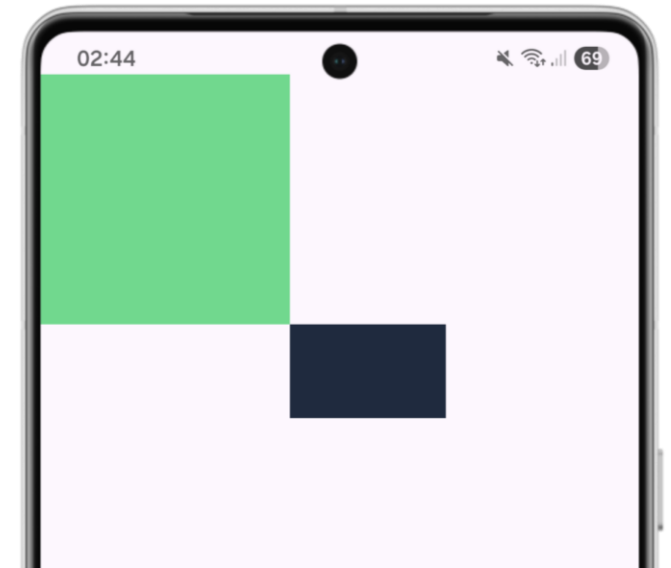
Constraint Layout



Top - Bottom



Start - End

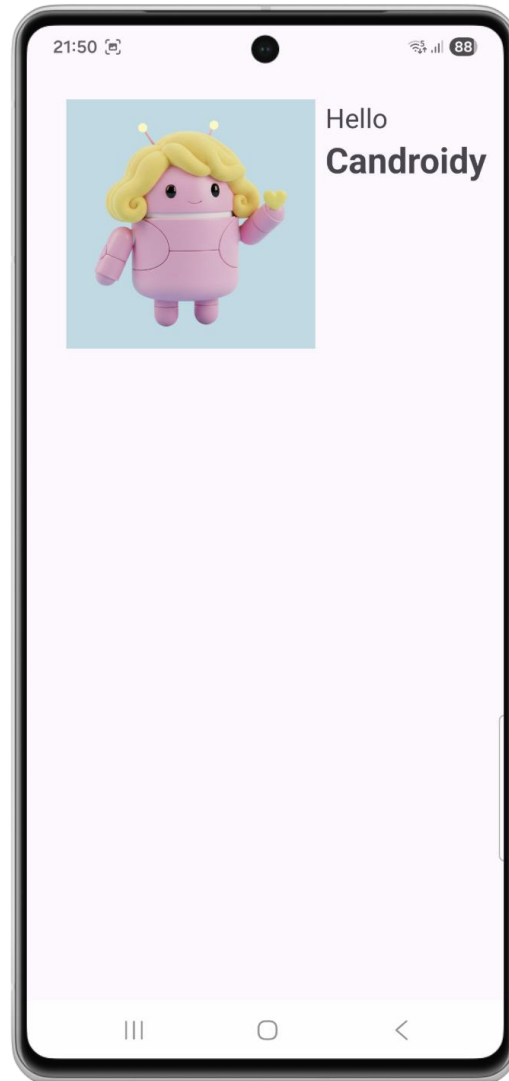


Constraint Layout (Notes)

🤖 Constraint Layout depends on the relation between views

🤖 Connect a view vertically & horizontally unless it is `match_parent`

Meet Candroidy (Training)



Constraint Layout Benefits

- 🤖 Helps you build Responsive UI
- 🤖 More flexible than Relative Layout
- 🤖 Improved performance
- 🤖 Better memory usage
- 🤖 Highly reliable

Constraint Layout (Part 1): <https://www.youtube.com/watch?v=NGuOouj3PE0>

Constraint Layout (Part 2): <https://www.youtube.com/watch?v=GJXBCRYOrpA>

Constraint Layout (Part 3): https://www.youtube.com/watch?v=mHameoXci_A

Choosing a View Group

Choose the ViewGroup that fits your needs without adding unnecessary complexity

- For example, if a `LinearLayout` can accomplish the task, there is no need to use a `ConstraintLayout`

In Summary

Choose your ViewGroup depending on the views arrangement on the screen

Better Performance

if your app calls `setText()` on a view object that wraps text, the view might need to resize (`wrap_content` is expensive).

A common reason for layout taking a long time is when hierarchies of `View` objects are nested within one another. Each nested layout object adds cost to the layout stage. The flatter your hierarchy, the less time it takes for the layout stage to complete.

Better UI

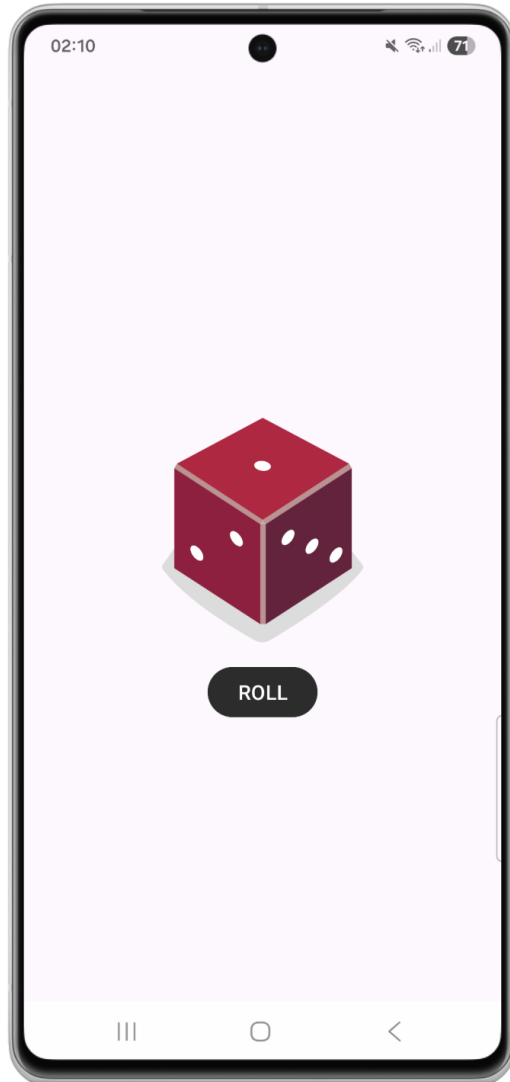
MaadCoder

```
app:layout_constraintBottom_toBottomOf="@id/maad_tv"
```

MaadCoder

```
app:layout_constraintBaseline_toBaselineOf="@id/maad_tv"
```

Exercise



1: <https://www.youtube.com/watch?v=nMTIcSw9rto>

2: <https://www.youtube.com/watch?v=o2BJbu11P-M>

3: <https://www.youtube.com/watch?v=zuZz0nGJQII>

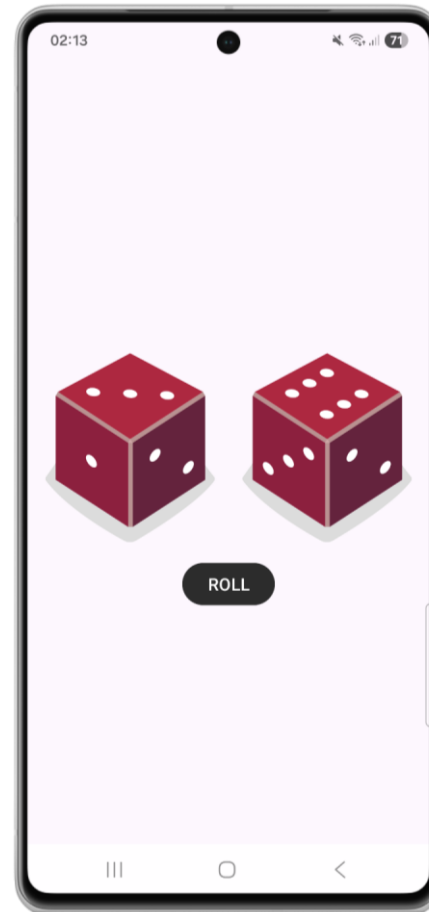
4: <https://www.youtube.com/watch?v=eX0yqUoQH6A>

5: <https://www.youtube.com/watch?v=S01Ehf8jjBw>

6: <https://www.youtube.com/watch?v=Es1MdieqG6g>

Challenge

Add another Dice to your application to simulate a real game



Submission Process

- 🤖 Upload your project to a **public** GitHub repository
- 🤖 Include a **readme** file explaining your project
- 🤖 Record a **video** of your application without showing the code, then share it in our group

Questions?



The End

